Multi-Sensor Fusion for Indoor Smartphone Localization with Attention Mechanism

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Abstract

Indoor localization has become an essential technology for navigation, asset tracking, and context-aware applications in environments where GPS signals are unavailable or unreliable. In this work, we present a multi-sensor fusion framework for indoor smartphone localization that integrates Inertial Measurement Unit (IMU) data with Wi-Fi Received Signal Strength Indicator (RSSI) measurements. IMU data provides high-frequency motion tracking, capturing short-term user movements, while Wi-Fi RSSI offers absolute positioning references from surrounding access points. To address the limitation of IMU (dead reckoning) and Wi-Fi RSSI (fluctuations and loss), we incorporate an attention mechanism that adaptively assigns higher weights to the most reliable input features. This allows the system to maintain accurate localization even when up to 90% of Wi-Fi measurements are missing. Experimental evaluations in realistic indoor settings demonstrate that our approach significantly reduces localization error compared to traditional sensor fusion.

Keywords

Indoor Localization, Multimodal Sensing, Sensor Fusion, Wi-Fi Fingerprinting, Location Tracking, Recurrent Neural Networks

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1 Introduction

Indoor localization has become a critical technology for navigation, asset tracking, and context-aware services in environments where GPS signals are unavailable or unreliable [19]. Its applications span diverse domains, including healthcare [11], retail, transportation, and smart buildings, driving demand for accurate and scalable positioning solutions [20]. Existing indoor localization methods, including Wi-Fi fingerprinting, ultra-wideband, and inertial dead reckoning, each face inherent challenges in real-world deployments. Wi-Fi-based approaches are prone to signal fluctuations and device variations [5, 13], while inertial methods gradually accumulate drift without external reference points [7, 15].

Recent studies have increasingly adopted multi-sensor fusion to mitigate these limitations. Work [1] integrated UWB, IMU, and visual-inertial odometry (VIO) using a sliding-window factor graph and an extended Kalman filter (EKF), achieving substantial accuracy improvements over single-sensor systems. However, this approach requires dedicated receiver hardware and precise hardware-level synchronization, which increases deployment cost and complexity. Wei et al proposed leveraging IMU sensors for building a pedestrian dead reckoning navigation system using LSTM network [12]. Work [9] proposed an xLSTM-based deep fusion framework combining UWB and IMU data with residual modules and attention mechanisms, achieving sub-0.2 m accuracy under both static and dynamic non-line-of-sight (NLOS) conditions. Nevertheless, this method demands a large amount of training data, which may limit its scalability in practice. Furthermore, the OWP-IMU dataset introduced by Zhang et al. Another work [16] introduced the OWP-IMU dataset, providing a benchmark for optical wireless positioning (OWP) fused with IMU data via EKF. While it demonstrates notable

accuracy gains under both line-of-sight and NLOS scenarios, the requirement for numerous infrared LEDs installed on ceilings significantly increases deployment and maintenance costs.

Despite recent progress, current sensor fusion approaches still face major challenges in indoor localization. Some methods rely on complex models to handle imbalanced or discontinuous sensor data, which increases system complexity and limits their deployment on resource-constrained devices. For example, work [3] applied convolutional neural networks (CNNs) to Wi-Fi fingerprints and achieved good accuracy, but their model could not effectively integrate IMU data, leading to drift during long trajectories. Work [6] explored transformer-based models that capture temporal dependencies, yet these approaches require heavy computation and large-scale training data, making them impractical for real-time smartphone applications. Another work [17] introduced a GAN-based framework to generate additional Wi-Fi samples for robustness and achieved a sufficient accuracy, but the framework doesn't support multi-rate sensor inputs such as IMU. These examples highlight that existing AI-based sensor fusion methods either demand complicated architectures or large training datasets, both of which limit scalability and real-world applicability.

In this paper, we propose an indoor mobile phone tracking and localization system that leverages two complementary sensing modalities. The first is Wi-Fi RSSI, collected from multiple Wi-Fi access points to provide coarse absolute position estimates. The second is inertial data from the smartphone's built-in IMU, which captures continuous motion and orientation changes. These data streams are fused using a framework enhanced with an attention mechanism, enabling the system to dynamically focus on the most reliable inputs. As a result, the proposed method can maintain adequate localization performance even when up to 90% of the Wi-Fi RSSI measurements are missing. Compared to above mentioned studies, this work makes the following contributions:

- **Lightweight attention-based fusion model**: We design an attention-based fusion framework that dynamically balances Wi-Fi RSSI and IMU inputs. This lightweight model improves localization accuracy while keeping computational complexity low, making it suitable for smartphone deployment.
- **Validation with real-world dataset**: The proposed approach is evaluated using a dataset collected from realistic indoor environments with natural signal fluctuations. Experimental results show consistent accuracy improvements compared to single-sensor and fixed-fusion baselines.

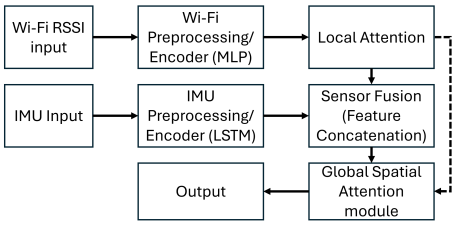


Figure 1: Our proposed multimodal deep neural network architecture two parallel single-modality extractors, a feature concatenation for sensor fusion with global/local attention mechanism.

- **Robustness under Wi-Fi signal loss**: Our method remains effective even when Wi-Fi data is sparse or unreliable. The attention mechanism enables the model to maintain reliable localization performance with up to 90% of RSSI measurements missing.
- **Public Dataset**: We aim to release the dataset and source code upon publication. The dataset contains both WiFi and IMU modalities, collected from two unique indoor office buildings at different geolocational places with ground truth location labels.

This paper is organized as follows: the proposed attention mechanism is presented in Section II. Data description and collection are given in Section III. Experimental Results and evaluation are presented in Section IV. Finally, conclusions are drawn in Section V.

2 Sensor Fusion Framework

We propose a multimodal data fusion–based positioning system that exploits strengths of both Wi-Fi RSSI and IMU measurements. While Wi-Fi RSSI provides absolute position references, it is often affected by signal instability, multipath effects, or temporary disconnections. To address these challenges, the fusion framework must remain robust even when Wi-Fi data is sparse or unreliable. To this end, we incorporate an attention mechanism that enables the model to dynamically assign higher weights to the most informative and reliable sensor inputs. This advanced strategy ensures consistent localization performance, even under severe Wi-Fi data loss or degradation.

The proposed overall architecture is shown in Fig 1. It takes Wi-Fi RSSI and IMU data as separate inputs, which are individually preprocessed and encoded using a multilayer perceptron (MLP) for Wi-Fi and a long short-term memory (LSTM) network for IMU. Inspire by the attention mechanism in vision processing [2], a local attention module is applied to the Wi-Fi features to emphasize reliable signal information before concatenating them with the IMU features in the sensor fusion layer. The fused features are further refined

through a Global Spatial Attention Module (GSAM), which assigns updated weights to feature maps and outputs the final position estimates. Full details are demonstrated in Table **??**.

2.1 IMU Data Processing

At each time step t, the LSTM updates its hidden state h_t and cell state c_t based on the input IMU vector x_t. Through the gating mechanism, the network selectively stores useful past information while incorporating relevant new input, allowing it to capture both short-term motion variations and long-term dependencies [10]. After processing the full sequence of IMU inputs, we obtain the final embedding vector by projecting the last hidden state h_T into a fixed-dimensional representation:

$$f_{IMU} = W_h h_T + b_h \tag{1}$$

where W_h is the projection matrix, b_h is the bias vector added to the projection, and $f_{IMU} \in \mathbb{R}^{128}$ is the temporal embedding feature vector that summarizes high-level representation of IMU features. In this work, a single-layer LSTM with 128 units has been used.

2.2 Wi-Fi RSSI Data Processing

To process the Wi-Fi RSSI measurements, we adopt a 3-layer MLP network that transforms the raw RSSI vector into a compact feature representation as stated in [4]. The input vector r contains the RSSI values from multiple access points, which often exhibit fluctuations and noise due to multipath effects. The MLP consists of three fully connected layers with 128 units each, where the first two layers use ReLU activation to introduce nonlinearity and robustness, and the final layer outputs a 128-dimensional embedding.

$$f_{wifi} = W_3 \times ReLU(W_2 \times ReLU(W_1 r + b_1) + b_2) + b_3 \tag{2}$$

where $ReLU$ indicates rectified linear unit and b_1, b_2, and b_3 are the bias for each layer respectively. This feature vector $f_{wifi} \in \mathbb{R}^{256}$ serves as the high-level representation of Wi-Fi features, which is then fused with the IMU representation in the attention-based fusion framework.

2.3 Feature Concatenation and Fusion

To integrate the information from both sensing modalities, we adopt the feature concatenation and fusion strategy [18]. The IMU feature f_{IMU} and the Wi-Fi RSSI feature f_{wifi} are first concatenated into a joint representation: $F = [f_{IMU}; f_{wifi}]$. This fused feature vector is then passed through a fully connected projection layer to obtain the final multimodal as

$$\hat{F} = \mathrm{ReLU}(W_f F + b_f) \in \mathbb{R}^{256} \tag{3}$$

The out feature $\hat{F}$ includes both IMU and Wi-Fi RSSI in a unified representation, which will be studied by attention mechanism.

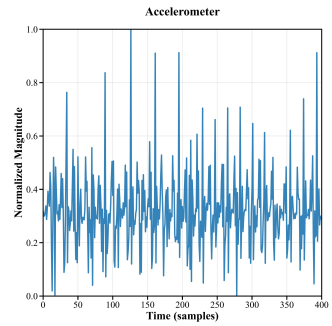


Figure 2: Raw IMU-Accelerometer sensor data visualization showing normalized magnitude over time of the accelerometer measurements. All data are normalized to the range [0,1] and represent the Euclidean norm of the three-axis sensor readings.

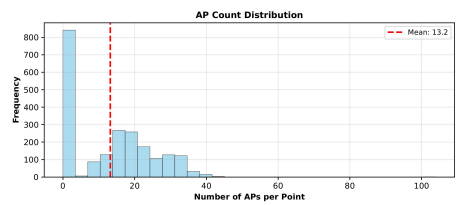


Figure 3: Statistic results of the access point coverage per location

2.4 Global Spatial Attention Module

In this work, GSAM has been used to enhance the fused feature representation by focusing the most informative spatial dimensions. Unlike simple concatenation or averaging, GSAM adaptively extract higher weights to reliable features while suppressing noisy or redundant ones. This mechanism is particularly useful for Wi-Fi RSSI data, which often suffers from fluctuations and disconnections. Considering an unstable scenario, like a crowded place, where the Wi-Fi signal is not always available and reliable, attention mechanism could help the AI model to identify the meaningful data. By refining the joint feature through global and local spatial attention, GSAM ensures more robust and accurate localization performance [8]. Specifically, the global attention captures long-range dependencies across the entire feature space, allowing the model to extract consistent patterns that span multiple access points. Meanwhile, the local attention focuses on fine-grained variations, highlighting short-term changes in RSSI or IMU signals that might be ignored.

Given a fused feature map F_l and a global feature map G_l, GSAM first aggregates global contextual information using two complementary pooling operations. The channel-wise average pooling F_l^{Avg} can be calculated as,

$$F_l^{Avg} = AvgPool(F_l) \qquad G_l^{Avg} = AvgPool(G_l) \tag{4}$$

Table 1: Multimodal Dataset Format

Time	Accelerator	AP0	AP1	...	APn	X	Y
T0	a*(0~999)*	null	-86	...	null	X*0*	Y*0*
T1	a*(999~1999)*	null	null	...	null	X*1*	Y*1*
T2	a*(1999~2999)*	-70	null	...	-65	X*2*	Y*2*
...	...	...	...	...	...	...	...
Tn	a*(n~n+999)*	null	null	...	null	X*n*	Y*n*

captures the overall distribution of feature activations, providing a smooth global context. In contrast, the channel-wise max pooling,

$$F_l^{Max} = MaxPool(F_l) \qquad G_l^{Max} = MaxPool(G_l) \quad (5)$$

It highlights the most dominant and discriminative responses, which provides that strong local signals are stored. These pooled features are combined and passed through a 1×1 convolution with sigmoid activation to generate an attention map:

$$M_s = \sigma\left(f_{1\times1}\left(\sigma\left(F_l^{\text{Avg}} + \hat{G}^{\text{Avg}}\right) + \sigma\left(F_l^{\text{Max}} + \hat{G}^{\text{Max}}\right)\right)\right), \quad (6)$$

where $f_{1\times1}$ denotes the convolution operator and σ is the sigmoid activation. The resulting attention map M_s assigns updated weights to different spatial locations, which are applied element-wise to the original feature map:

$$\hat{F}_l = M_s \otimes F_l, \quad (7)$$

where $\otimes$ denotes element-wise multiplication between the attention map M_s and the original feature map F_l. This approach ensures that reliable Wi-Fi data could be selected. Consequently, GSAM could enhance localization accuracy under real-world indoor conditions.

2.5 Joint Classifier

The refined feature representation $\hat{F}_l$ is then passed into a joint classifier for final localization. The first fully connected layer contains 128 units with ReLU activation, which projects the fused features into a compact and discriminative space. The second fully connected layer has 2 output units with ReLU activation, corresponding to the predicted 2D location coordinates in X-axis and Y-axis.

2.6 Classifier Training

For the training process, the model was configured with a batch size of 100 and optimized using the root mean square propagation algorithm with a learning rate of 0.0001. The loss function employed was mean squared error (MSE), ensuring precise regression performance for location prediction. The network was trained for 200 epochs, with a hidden size of 128 to balance computational efficiency and representation capability.

3 Experimental Data

To construct the dataset for the multi-modal indoor localization task, we developed a dedicated Android application for data collection, where the dataset was initially used in [14]

During data collection, ground truth labels were exchanged between two devices. In the initial campaign, participants walked along a corridor at varying speeds, starting from one corner of the building, following a circular route, and returning to the origin. Ground truth positions were recorded intermittently; to achieve finer location granularity, additional positions were interpolated between consecutive recorded points under the assumption of locally constant walking speed. For each round of data collection, two synchronized datasets were constructed: Set 1 contained inertial sensor readings and WiFi scans captured by phone 1 during the walk; Set 2 comprised ground truth location labels, which were manually input on phone 2 by entering the latitude and longitude of phone 1 at key positions such as corners and doorways.

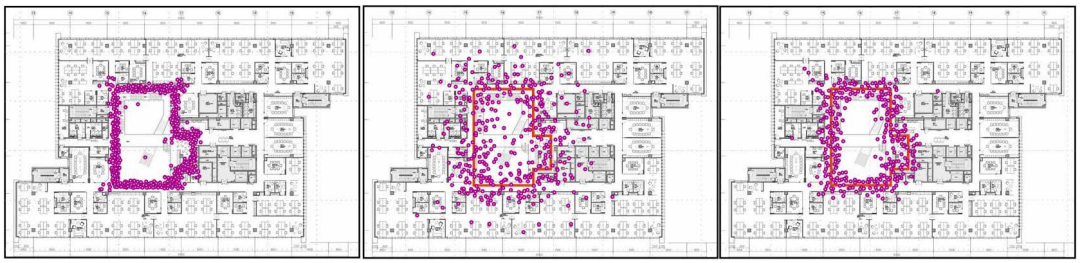
The application leverages Android APIs to continuously record IMU signals and WiFi RSSI in the background using an event-based sampling mechanism with a fixed timestep. In addition, an interactive map interface aligned with the Google Maps API enables the user to annotate the location of the ground truth by clicking on the screen to obtain latitude and longitude coordinates.

During data acquisition, the logging function is simultaneously activated on two smartphones. One device is placed in the user's pocket to capture IMU signatures in natural motion, while the other device is used to manually annotate the position of ground truth when passing characteristic points such as corners, elevators, etc. An example of the collected IMU data is shown in Figure 2.

The accelerometer measurement is derived by calculating the magnitude value on the three orthogonal axes based on the raw three-dimensional samplings, where acc$\{x, y, z\}$ are the measurement on each Cartesian axes.

$$Acc_{magnitude} = \sqrt{acc_x^2 + acc_y^2 + acc_z^2}$$

For this work, the dataset comprises 24,450 inertial sensor movement samples and 25,541 WiFi scans from a total of 102 WiFi access points. Data was collected in total of over 6 hours, using three different smartphones. Table 1 illustrates the streamlined dataset structure. At each millisecond timestep, the system records a sequence of accelerometer readings within a one-second window, along with the WiFi RSS fingerprints available at that moment. If no WiFi data is detected, a *null* value is used to indicate missing entries. During data collection, ground truth positions are manually annotated when passing notable locations such as corners,



(a) (b) (c)

Figure 4: Footpath Visualization of Position Estimation with Different Data Configurations: (a) Visualization using complete datasets for IMU, and Wi-Fi data. (b) Visualization using 10% of Wi-Fi data combined with full IMU data. (c) Visualization using 10% of Wi-Fi data, full IMU data, and incorporating an attention mechanism.

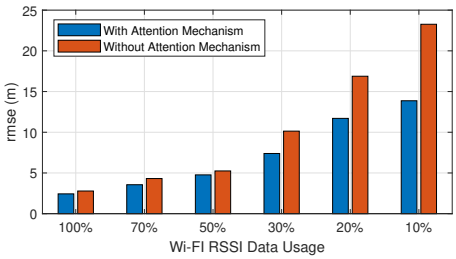


Figure 5: Comparison of rmse between with and without attention mechanism

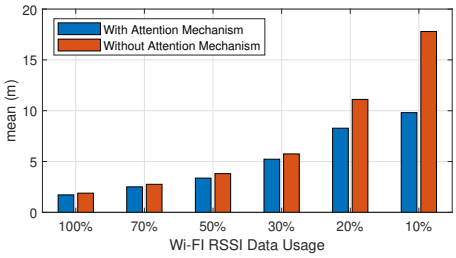


Figure 6: Comparison of mean error between with and without attention mechanism

elevators, or staircases. For all other timesteps, location labels are generated by interpolating between these reference points, resulting in a fully labelled time-series dataset.

Figure 3 represents the statistic results demonstrating the bias distribution of the scanned WiFi access points per Fingerprinting location, where over 50% of the location has no WiFi signal coverage. It indicates the signal availability varied significantly in the data collection scenario, which is commonly happened in real-life situations.

4 Experimental Results

Fig 4 illustrates the footpath visualization of position estimation under three different data configurations. In Fig 4(a), the trajectory generated using complete datasets for both IMU and Wi-Fi RSSI closely follows the actual walking path, indicating high localization accuracy. In Fig 4(b), when only 10% of the Wi-Fi data is available and combined with full IMU data, the estimated path deviates more significantly from the ground truth, reflecting reduced positioning accuracy due to limited Wi-Fi references. In Fig 4(c), our proposed attention-enhanced fusion method is applied under the same limited Wi-Fi RSSI data. As it can be seen, the resulting trajectory shows notable improvements compared to Fig 4(b), and shows closely performance in Fig 4(a). Moreover, attention mechanism still performs well even there are 100% of Wi-Fi RSSI data.

Fig 5 and 6 compare the localization performance with and without the attention mechanism under different levels of Wi-Fi RSSI availability reduced from 100% to 10%. As shown in Fig 5, the RMSE of the proposed method remains consistently lower than the baseline (which uses same network but without attention mechanism) across all scenarios. This advantage has became more significant when the proportion of Wi-Fi data decreases. It reaches RMSE of 13.87 with attention mechanism and 23.26 without attention mechanism. Similarly, Fig 6 demonstrates that the mean localization error is also reduced when using the attention mechanism. It is observed that the mean error increased to 9.81m with attention mechanism compared to 17.79m without attention mechanism, almost 55% improved. These results confirm that our proposed attention mechanism can significant maintaining accuracy even under severe Wi-Fi signal sparsity.

Fig 7 shows the Cumulative Distribution Function (CDF) of localization error under different Wi-Fi RSSI usage ratios. With 100% RSSI data, about 90% of the errors are below 2 meters, demonstrating the highest accuracy. When only 50% of RSSI data is used, around 80% of the errors remain within 5 meters, while the error distribution becomes noticeably wider. At the worst case of 10% RSSI availability, fewer than

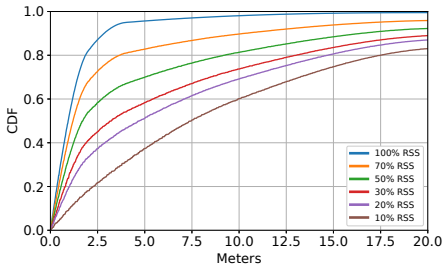


Figure 7: CDF of localization error under different Wi-Fi RSSI ratios.

40% of the errors are under 5 meters, and the rest error extends beyond 15 meters. This showcases the degradation in performance under when Wi-Fi signal is significantly sparsity.

5 Conclusion

This paper presented a multi-sensor fusion framework for indoor smartphone localization that integrates Wi-Fi RSSI and IMU data through an attention mechanism. The proposed method addresses the limitations of each modality, mitigating IMU drift with Wi-Fi references while reducing sensitivity to Wi-Fi fluctuations and data loss using inertial cues. By incorporating both local and global attention modules, the system dynamically emphasizes reliable features, enabling robust and accurate localization even when up to 90% of Wi-Fi data is missing. Experimental evaluations on a real-world dataset demonstrated continuous improvements over traditional fusion and attention mechanism. The results showed significant reductions in both RMSE and mean error, and the CDF analysis confirmed the effectiveness of the method across different Wi-Fi availability.

Future work will focus on further enhancing the adaptability and scalability of the proposed framework. One direction is to explore transfer learning and domain adaptation techniques to generalize the model across different locations and device types without extensive retraining. Another direction is to optimize the model for low-power operation which is essential for real-time applications, to enable the sensor fusion technique on smartphone platform.

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